Virtual Mouse

Emanuel Charro, 862158963
JoseAngel Pulido, 862398468
EE146, W26

Problem & Related Work

Purpose

- Develop a touchless mouse control system using a webcam.
- Track hand gestures to move the cursor and perform mouse clicks
- Use computer vision and MediaPipe hand tracking to recognize finger positions

Scope

- Use webcam as real time input
- Detect hand landmarks and fingertip positions
- Map finger movement to screen cursor movement
- Detect pinch gestures for mouse clicks

Related Work

Many computer vision systems rely on features extraction and object analysis to interpret images. Our projects applies these ideas by detecting hand landmarks and classifying gestures for real time mouse control.

Technical Approach I

Pipeline

Step 1

- Build the custom gesture recognizer model using a dataset of ~700 images of hand gestures.

Step 2

- Evaluate the model by deriving the confusion matrix, precision, and f1 score.

Step 3

- With metrics at +74%, we implement the virtual mouse.

Techniques Used

`gesture_recognizer.Dataset.from_folder`

- Automated Feature Extraction

`gesture_recognizer.GestureRecognizer.create`

- Supervised Deep Learning

`HandDataPreprocessingParams()`

- Interpolation & Coordinate Mapping for scale-invariance.

- Statistical Evaluation
- Supervised learning
- Decision thresholding

- Convolutional Neural Networks (CNN)
- Mirroring/Color Space Conversion
- Classification
- Linear Interpolation
- Smoothing

Technical Approach II

Methodology

Label images into gesture classes:
- Point
- Left-click
- Right-click

Train a gesture recognition model using MediaPipe Model Maker

- Evaluate the trained model using test images
- Measure classification performance using accuracy and confusion matrix

Normalization

Hand landmarks are normalized coordinates between 0 and 1. This allows the model to work regardless of the distance from the camera

- **Smoothing filter** to reduce noise between video frames.
- **Frame flipping** to align the camera's perspective with the user's sense of direction.
- **Linear interpolation** to map the camera's detected points to the monitor's resolution.

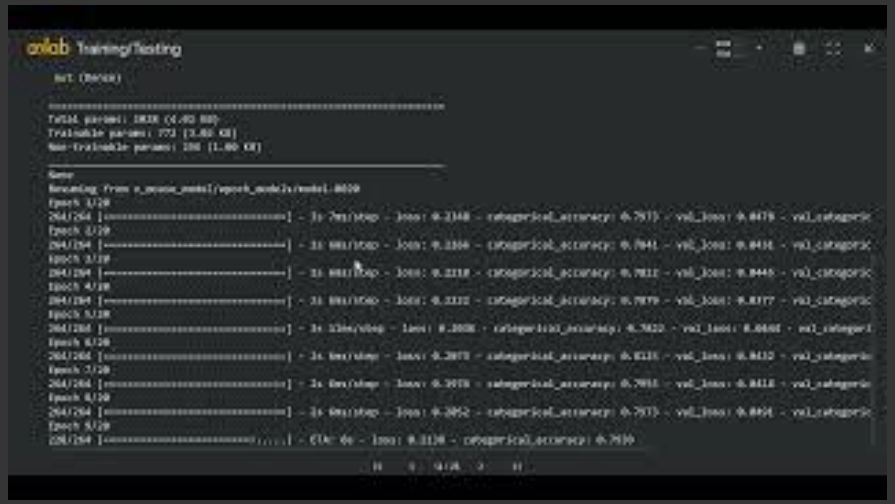
Feasibility

MediaPipe provides efficient real-time hand tracking. Real time performance makes it practical for interactive applications

With the use of deep learning techniques such as
- Coordinate regression
- Backpropagation
- Gradient descent,

using MediaPipe as the backbone of this project significantly improves its performance

Demonstration



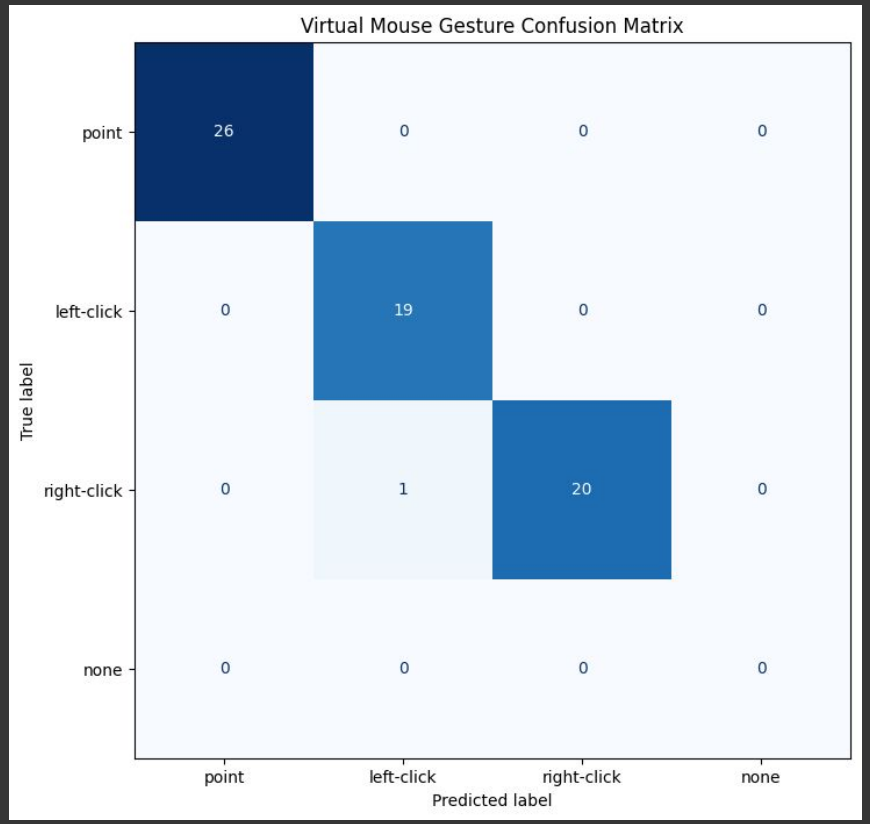
Model Construction & Metric Evaluation

Virtual Mouse Test

Results I

The result show the gesture recognition performs very well with ~98% accuracy. Most gestures were classified correctly all predictions appearing along the diagonal. The point gesture was detected perfectly. While the left-click and right-click gestures had only minor misclassification. Overall, the system demonstrates reliable gesture recognition for controlling the virtual mouse.

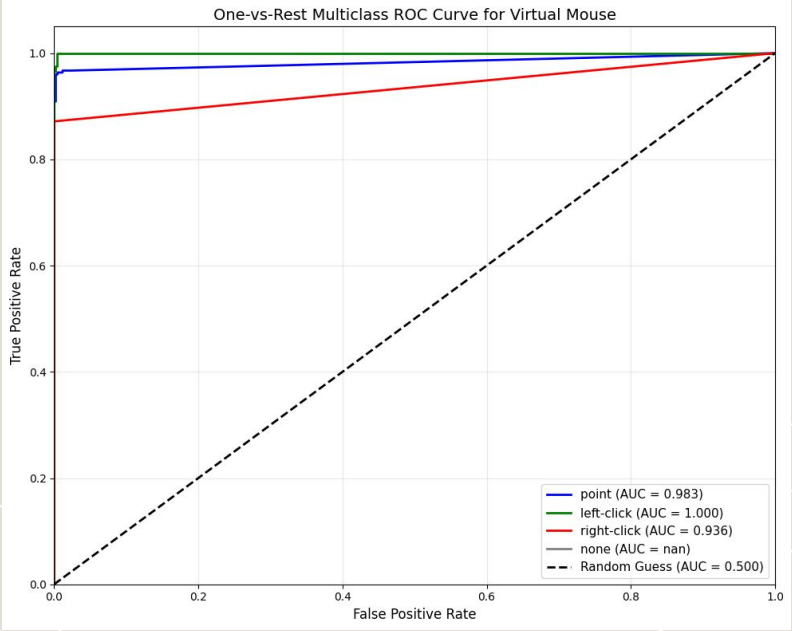
```
============================================================
Model Classification Summary
============================================================
              precision    recall  f1-score   support

       point       1.00      1.00      1.00        26
  left-click       0.95      1.00      0.97        19
 right-click       1.00      0.95      0.98        21
        none       0.00      0.00      0.00         0

    accuracy                           0.98        66
   macro avg       0.74      0.74      0.74        66
weighted avg       0.99      0.98      0.98        66
```



Results II

Evaluated Metrics

- ROC curve used to analyze how well the model distinguishes between gesture classes.
- AUC values close to 1.0 indicate strong gesture recognition performance



Conclusion

Current Stage	Currently the virtual mouse only includes functionality for the mouse cursor, left-clicks, and right-clicks.
Future Work	Introducing scroll functionality using the index and middle finger. The image difference nad motion vector between the neutral state and a “scroll down” or “scroll up” state indicates the direction of the scroll.
Significance	A virtual finger-controlled mouse via camera can be vital for use in medical settings where hygiene is critical or for individuals who lack proper hand coordination.

Thank you